

Zaby — Internal Services Knowledgebase

Product, Infrastructure & Service Definitions

1. Overview

Zaby is an AI-native operational infrastructure platform designed to help businesses deploy autonomous AI systems capable of execution, workflow automation, operational intelligence, and multimodal interaction.

This document defines:

- internal service architecture
- platform capabilities
- infrastructure layers
- runtime systems
- operational abstractions
- deployment models
- enterprise service categories

This document is intended for:

- internal product teams
- engineering teams
- design teams
- solution architects
- GTM teams
- implementation partners
- enterprise onboarding

2. Platform Philosophy

Most AI products today are designed around conversation.

Zaby is designed around execution.

The core architectural philosophy is:

“AI systems should operate like digital operational infrastructure.”

That means agents should:

- execute tasks
- interact with software
- operate continuously
- retain memory
- collaborate with humans
- coordinate with other agents
- function inside enterprise operational environments

The platform is intentionally designed to move beyond:

- static copilots
 - prompt wrappers
 - isolated automation tools
 - narrow chatbot systems
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3. Core Service Architecture

Zaby consists of five major service layers:

1. Agent Squad
2. Open Agents
3. Agentic Workflows
4. Agent Memory
5. AI SaaS Workspace

These layers are interconnected through:

- execution runtime
 - orchestration systems
 - scheduling infrastructure
 - observability
 - permissions
 - memory systems
 - multimodal interaction layers
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4. Agent Squad

Definition

Agent Squad is the operational AI employee layer of the Zaby platform.

It enables organizations to deploy specialized autonomous AI employees capable of performing operational tasks using:

- browser interaction
 - software operation
 - workflow execution
 - reasoning systems
 - tool integrations
 - computer-use infrastructure
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Strategic Purpose

The purpose of Agent Squad is to provide execution-capable digital employees rather than passive conversational assistants.

These agents are designed to function inside business operational environments.

Operational Characteristics

Core Characteristics

- autonomous execution
 - long-running tasks
 - contextual reasoning
 - operational continuity
 - multimodal interaction
 - enterprise permission handling
 - task persistence
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Supported Execution Modes

Browser Interaction

Agents can:

- navigate web interfaces
- interact with forms
- perform browser-based operations
- collect information
- operate dashboards

Computer Use

Agents can:

- interact with desktop applications
- execute operational workflows
- coordinate application-level actions
- perform UI-based automation

API Execution

Agents can:

- call APIs
- process structured responses
- integrate with internal systems
- trigger external actions

Workflow Coordination

Agents can:

- coordinate subtasks
- delegate work
- collaborate with other agents
- escalate tasks to humans

Example Agent Types

Operations Agents

- operational monitoring
- dashboard verification
- infrastructure checks

- process validation

QA Agents

- regression testing
- UI validation
- workflow testing
- automation verification

Support Agents

- ticket resolution
- customer interaction
- escalation handling
- support triage

Recruitment Agents

- candidate screening
- interview coordination
- assessment management
- workflow scheduling

Research Agents

- information gathering
- market analysis
- document summarization
- competitive intelligence

Technical Components

Runtime Engine

Responsible for:

- execution lifecycle
- agent state management
- task orchestration
- runtime isolation

Tool Layer

Provides:

- API connectivity
- browser actions

- external integrations
- operational tooling

Observation Layer

Tracks:

- task execution
- logs
- failures
- interaction history
- workflow states

Permission Layer

Controls:

- access management
 - environment permissions
 - operational restrictions
 - role boundaries
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5. Open Agents

Definition

Open Agents is the customizable multimodal agent creation layer of the platform.

It enables organizations to build domain-specific AI agents for customer-facing and internal operational workflows.

Strategic Purpose

The goal of Open Agents is flexibility.

Unlike Agent Squad, which focuses on predefined operational AI employees, Open Agents enables businesses to create highly customized AI systems for specific workflows, channels, and industries.

Supported Modalities

Text

- chat interfaces
- messaging systems
- support conversations
- internal collaboration

Voice

- inbound call handling
- voice assistants
- appointment booking
- conversational operations

Vision

- image understanding
- document analysis
- visual workflows
- UI interpretation

Browser Interaction

- web actions
- portal interaction
- operational navigation

API Interaction

- backend connectivity
- external system integrations
- workflow triggering

Example Open Agent Deployments

Ecommerce Agent

Capabilities:

- order support
- product recommendations
- return workflows
- customer FAQs

Restaurant Voice Agent

Capabilities:

- reservation handling
- call answering
- ordering assistance
- scheduling

Real Estate Assistant

Capabilities:

- lead qualification
- property recommendations
- appointment coordination

AI Tutor

Capabilities:

- educational assistance
- assessment support
- interactive learning

Deployment Channels

- websites
- WhatsApp
- Slack
- Discord
- internal dashboards
- mobile applications
- voice systems
- phone systems
- support portals

Technical Components

Agent Builder

Responsible for:

- agent configuration

- workflow setup
- integration management
- deployment controls

Multimodal Runtime

Handles:

- voice processing
- image interpretation
- text reasoning
- interaction coordination

Tool Integration Layer

Provides:

- CRM integrations
 - scheduling integrations
 - payment integrations
 - support tooling
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6. Agentic Workflows

Definition

Agentic Workflows is the orchestration and autonomous process execution layer of the platform.

It enables businesses to automate repetitive, operational, scheduled, and reasoning-based workflows.

Strategic Purpose

Traditional automation systems rely heavily on static logic.

Agentic Workflows introduces:

- reasoning
- adaptation
- conditional execution
- operational intelligence

into workflow systems.

Supported Workflow Types

Scheduled Workflows

Examples:

- daily audits
- reporting
- infrastructure checks
- monitoring tasks

Event-Driven Workflows

Examples:

- customer triggers
- system alerts
- workflow escalations
- operational events

Multi-Agent Workflows

Examples:

- collaborative execution
- sequential task handling
- distributed workflows

Human Approval Workflows

Examples:

- escalation systems
- review processes
- approval pipelines

Core Workflow Capabilities

- scheduling
- retries
- failure recovery
- observability
- event handling
- notifications
- approvals
- branching logic

- agent chaining
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Example Operational Use Cases

QA Automation

- regression testing
- deployment verification
- browser flow testing

Compliance Operations

- operational audits
- reporting validation
- monitoring

Infrastructure Monitoring

- uptime checks
- incident analysis
- operational validation

Content Operations

- publishing workflows
 - moderation pipelines
 - generation validation
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Technical Infrastructure

Workflow Engine

Responsible for:

- execution control
- orchestration
- scheduling
- retries
- branching

Event Bus

Handles:

- system events

- workflow triggers
- operational notifications

Execution Tracker

Tracks:

- workflow states
 - logs
 - failures
 - execution metrics
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7. Agent Memory

Definition

Agent Memory is the persistent intelligence infrastructure layer for AI systems.

It enables agents to retain:

- operational knowledge
- user context
- workflow state
- historical interactions
- organizational information

across sessions and environments.

Strategic Purpose

Most AI systems are stateless.

Agent Memory enables continuity.

The goal is to provide long-term contextual intelligence for autonomous systems.

Memory Types

Conversational Memory

Stores:

- user interactions
- preferences
- historical conversations

Operational Memory

Stores:

- workflow state
- execution history
- operational context

Knowledge Memory

Stores:

- organizational documents
- business knowledge
- reference materials

Shared Memory

Allows:

- multi-agent coordination
- shared operational context
- collaborative workflows

Core Capabilities

- semantic retrieval
 - vector storage
 - structured memory
 - context compression
 - long-term persistence
 - permission-aware retrieval
 - retrieval APIs
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Example Use Cases

Personalized Support

Agents remember:

- previous issues
- customer preferences
- historical interactions

Long-Running Workflows

Agents maintain:

- operational state
- execution continuity
- process memory

Enterprise Knowledge Systems

Agents access:

- company documents
 - internal procedures
 - operational guidelines
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Technical Components

Memory Store

Handles:

- storage
- indexing
- retrieval
- persistence

Retrieval Engine

Responsible for:

- semantic search
- contextual recall
- ranking
- memory filtering

Compression Layer

Optimizes:

- context size
 - retrieval efficiency
 - operational continuity
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8. AI SaaS Workspace

Definition

AI SaaS Workspace is the application-layer operational environment of the platform.

It provides prebuilt AI-native business workspaces with integrated:

- agents
 - workflows
 - analytics
 - collaboration systems
 - operational tooling
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Strategic Purpose

The objective is to accelerate deployment of operational AI systems.

Instead of requiring businesses to build infrastructure from scratch, AI SaaS Workspace provides production-ready operational environments.

Example Workspace Types

Hiring Workspace

Capabilities:

- candidate screening
- assessments
- interview coordination
- recruiter workflows

Assessment Platform

Capabilities:

- evaluation systems
- automated scoring
- workflow automation

Content Studio

Capabilities:

- content pipelines
- generation workflows
- collaboration
- review systems

Support Operations Center

Capabilities:

- ticket management
 - AI-assisted workflows
 - escalation handling
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Workspace Components

Embedded Agents

Provide:

- task automation
- workflow execution
- assistance
- operational support

Analytics Layer

Provides:

- operational visibility
- workflow metrics
- execution insights

Collaboration Systems

Enable:

- team coordination
- approval systems
- operational management

Workflow Systems

Handle:

- operational automation
 - process execution
 - task orchestration
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9. Shared Platform Infrastructure

All Zaby services operate on top of shared infrastructure systems.

Runtime Infrastructure

Responsible for:

- execution environments
 - agent isolation
 - scalability
 - concurrency
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Scheduling Infrastructure

Handles:

- recurring workflows
 - delayed execution
 - operational timing
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Observability Layer

Provides:

- logs
 - metrics
 - tracing
 - execution visibility
 - debugging
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Security Infrastructure

Includes:

- RBAC permissions
 - environment isolation
 - audit logs
 - secure integrations
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Integration Layer

Supports:

- APIs
 - CRMs
 - databases
 - operational tooling
 - enterprise systems
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10. Enterprise Deployment Models

Cloud Deployment

Managed infrastructure hosted by Zaby.

Dedicated Enterprise Deployment

Isolated enterprise infrastructure.

Self-Hosted Deployment

Customer-managed infrastructure environments.

11. Internal Strategic Positioning

The company should avoid internally positioning Zaby as:

- chatbot platform
- AI wrapper
- automation tool
- no-code assistant builder

The strategic positioning should remain:

Operational AI Infrastructure

and

AI Workforce Platform

12. Long-Term Vision

The long-term platform vision is to build:

- AI operational systems
- autonomous workforce infrastructure
- enterprise AI runtime layers
- multimodal execution systems
- collaborative human + AI operational environments

The focus is not replacing humans.

The focus is enabling organizations to scale operational execution through AI-native systems.

13. Internal Messaging Principles

All internal and external messaging should prioritize:

- execution
- reliability
- operational intelligence
- enterprise readiness
- infrastructure quality
- multimodal capability
- autonomy with control

Avoid:

- AGI claims
- hype-heavy messaging
- sci-fi positioning
- unrealistic automation promises
- “replace your employees” narratives

The positioning should remain:

calm, credible, operational, technical, and enterprise-capable.